

Haplotype Block Partitioning and TagSNP Selection with MapReduce framework

Che-Lun Hung

Department of Computer Science and Communication Engineering
Providence University
Taichung, Taiwan
clhung@pu.edu.tw

Yaw-Ling Lin

Department of Computer Science and Information Engineering
Providence University
Taichung, Taiwan
yllin@pu.edu.tw

Guan-Jie Hua

Department of Computer Science and Information Engineering
Providence University
Taichung, Taiwan
gt758215@gmail.com

Kuan-Ching Li

Department of Computer Science and Information Engineering
Providence University
Taichung, Taiwan
kuancli@pu.edu.tw

Abstract

SNPs are fundamental roles for various applications including medical diagnostic and drug design. They also provide the highest-resolution genetic fingerprint for identifying disease associations and human features. Genetic variants that are near each other tend to be inherited together; these regions of linked variants are known as haplotypes. Recently, genetics researches reveal that SNPs within certain haplotype blocks induce only a few distinct common haplotypes in the majority of the population. The existence of haplotype block structure has serious implications for association-based methods for the mapping of disease genes.

We proposed the method in investigating several efficient combinatorial algorithms related to selecting interesting haplotype blocks under different diversity functions that generalizes many previous results in the literatures. However, the proposed method is computation-consuming. This paper adopts approach using the MapReduce paradigm to parallelize tools and manage their execution. The experiment shows that the map/reduce-paralleled from the original sequential combinatorial algorithm performs well on the real-world data obtained in from the HapMap data set; the computation efficiency can be effectively

improved proportional to the number of processors being used.

1. Introduction

Genome-wide association studies based on *linkage disequilibrium (LD)* offer a promising approach to detect genetic variation responsible for common human diseases. *Single nucleotide polymorphisms (SNPs)* are promising markers for disease association researches because of their high abundance along the human genome, the low mutation rate and accessibility to high-throughput genotyping. A SNP refers to the existence of two specific nucleotides at a single locus in a population. Haplotype can be defined as an asset of SNPs on a single chromosome that are associated and inherited as a unit. Recently, haplotype analysis has been successfully applied to identify DNA variations that are relevant to several common and complex diseases [1-6]. Many studies suggest that human genome maybe arranged into block structure, in which SNPs are relevant and only a small number of SNPs are sufficient to obtain most of haplotype structures, called tagSNP [7-13].

Several approaches have been suggested for defining block structure from which some are more commonly used. Four main criteria for haplotype block partitioning are based

on haplotype diversity, LD, four gamete test and information complexity. In diversity-based methods [9, 14, 15], a *block* is defined as a region, in which a certain percentage of haplotypes are *common haplotypes*, haplotypes which are represented more than certain percent in the population. In LD-based methods [8, 16], a block is defined as a region with high pair-wise LD within block and low pair-wise LD between blocks. In methods based on four gamete test [17, 18], a block is defined as a recombination-free region of consecutive SNPs. However, existing approaches that address these issues are generally complicated and computationally intensive.

Hadoop [19] is a software framework intended to support data-intensive distributed applications. It is able to process petabytes of data with thousands of nodes. Hadoop supports MapReduce programming model [20] for writing applications that process large data set in parallel on Cloud Computing environment. The advantage of MapReduce is that it allows for distributed computing of the map and reduction operations. Each map operation is independent of the others and all maps can perform the tasks in parallel. In practice, the number of the map is limited by the data source and/or the number of CPUs near that data. Similarly, a set of reducers can perform the reduce operations. All outputs of the map operations which share the same key are presented to the same reducer, at the same time. In addition, one of the important benefits to use Hadoop to develop the applications is due to its high degree of fault tolerance. Even when running jobs on a large cluster where individual nodes or network components may experience high rates of failure, Hadoop can guide jobs toward a successful completion. Many applications of bioinformatics are often computation-consuming; sometimes it needs weeks or months to complete the jobs. The traditional parallel models, such as MPI, OpenMP and Multi-thread, are not suitable to such applications, where a fault occurred in some nodes leads the entire application into total failure. In these situations, the Hadoop platform are considered as a much better solution for these real-world applications. Recently, Hadoop has been applied in various domains in bioinformatics [21, 22].

In this paper, we propose parallel diversity-based haplotype block selection algorithms on Hadoop MapReduce framework. In particular, the map-stage is utilized to calculate the diversity of each block, while the reduce-stage is utilized to locate the blocks. As these experimental results indicate that these proposed algorithms are significantly faster than the corresponding sequential algorithms as the number of Map operations increased.

2. Method and Materials

C. Haplotype block partitioning

A SNP (Single Nucleotide Polymorphisms) is defined as a position in a chromosome where each one of two (or more) specific nucleotides is observed in at least 10% of the population [9]. The nucleotides involved in a SNP are called alleles. A SNP is said *biallelic* if it has only two different alleles. Almost all SNP are biallelic and we will consider exclusively biallelic SNP in this paper.

Diversity functions

Abstractly, the input to the haplotype blocking problem consists of m haplotype vectors. Each position in a vector is associated with a site of interest on the chromosome. Usually, the position in the haplotype vector has a value of 0 if it is the *major allele* or 1 if *minor allele*.

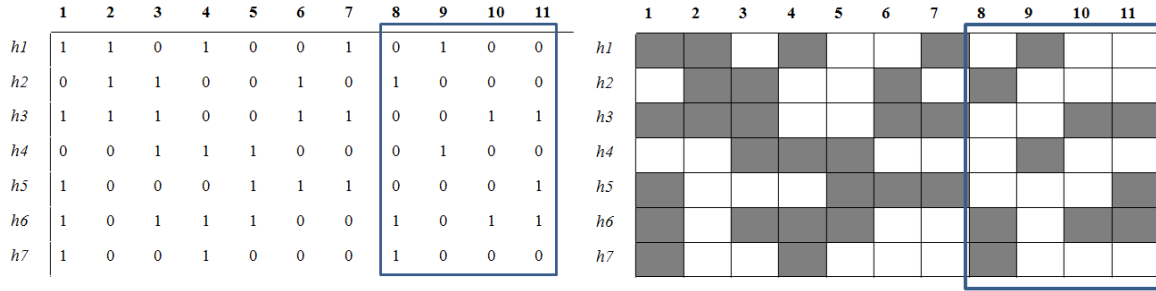
Let the haplotype matrix A be an $m \times n$ matrix of m observations over n markers (sites). We refer to the j th allele of observation i by A_{ij} . For simplicity, we assume that $A_{ij} \in \{0, 1\}$. A block, or marker interval, $[j, k] = \{j, j+1, \dots, k\}$ is defined by two marker indices $1 \leq j \leq k \leq n$. A *segmentation* is a set of non-overlapping non-empty marker intervals. A segmentation is full if the union of the intervals is $[1, n]$. The data matrix limited to interval $[j, k]$ is denoted by $M(j, k)$; the values of the i -th observation are denoted by $M(i, j, k)$, a binary string of length $k - j + 1$. Figure 1 illustrates an example of a 7×11 haplotype matrix.

Given an interval $[j, k]$, a *diversity function*, $\delta : [j, k] \rightarrow (j, k) \in R$ is an evaluation function measuring the diversity of the submatrix $M(j, k)$. We say an interval $[j', k']$ is a subinterval of $[j, k]$, written $[j', k'] \subset [j, k]$, if $j \leq j'$ and $k' \leq k$. Note that δ -function is a monotonic non-decreasing function from $[1..n, 1..n]$ to the unit real interval $[0, 1]$; that is, $0 \leq (\delta(j', k') \leq \delta(j, k) \leq 1$ whenever $[j', k'] \subset [j, k]$.

Given an input set of n haplotype vectors, a solution to the *Haplotype Block Selection (HBS)* Problem is a segmentation of marker intervals, revealing these non-overlapped haplotype blocks of interest in the chromosome. In our previous work [23], we have proposed several efficient algorithms related to selecting interesting haplotype blocks under different evaluation (diversity) functions that generalizes many previous results in the literatures [8, 9, 12, 13, 16, 17, 18, 24, 25, 26]. In this paper, we focus on finding common haplotypes.

Common Haplotypes

Two haplotypes are said to be *compatible* if the alleles are identical at all loci for which there are no missing data; otherwise the two haplotypes are said to be *incompatible*. As in Patil *et al.*, [9] we define the ambiguous haplotypes as those haplotypes compatible with at least two haplotypes that are themselves incompatible. It should be noted that when there are no missing data, all of the haplotypes are unambiguous. We define the common haplotypes as those

Figure 1. A haplotype matrix B and its corresponding submatrix $M(8, 11)$.

haplotypes that are represented more than once in a block. The haplotypes are called *singleton* if they are not compatible with any others.

We are mainly interested in the common haplotypes. Therefore we require that, in the final block partition, a significant fraction of the haplotypes in each block is common haplotypes. Patil et al. [9] require that at least $\alpha = 70\%$, 80% , and 90% , respectively, of the unambiguous haplotypes appear more than once. The parameter, α , is also referred to as the *coverage* of common haplotypes in a block. Ambiguous haplotypes are not included in calculating percent coverage. The coverage of block B can be mathematically formulated as a form of diversity:

$$\delta_s(B) = 1 - \frac{U}{C} = \frac{U}{S} \quad (1)$$

Here U denotes the number of unambiguous haplotypes, C denotes the number of common haplotypes, and S denotes the number of singleton haplotypes. For example, the submatrix $M(8, 11)$ of Fig. 1 can be viewed as a sample $S = \{0001, 0011, 0100, 1000, 1011\}$. It follows that $\delta(S) = \delta(M(8, 11)) = 1 - 4/7 = 3/7$.

TagSNPs Selection

For each block, we want to minimize the number of SNPs that uniquely distinguish at least 80% (the α parameter) of the unambiguous haplotypes in the block. Those SNPs can be interpreted as a signature of the haplotype block partition. They are referred to as tagSNPs that are able to capture most of the haplotype diversity, and therefore, could potentially capture most of the information for association between a trait and the marker loci [28].

Our strategy for selecting the tagSNPs in haplotype blocks is as the following. First, the common haplotypes are grouped into k distinct patterns in each block. After the missing data are assigned, as explained in the next subsection, we decide the least number of groups needed such that haplotypes in these groups contain at least 80% (α) of the unambiguous haplotypes in the block. Finally, we select a loci set which consists of the minimum number of SNPs on the haplotypes such that each pattern can be uniquely

distinguish. Exhaustive searching methods are used very efficiently here since the number of tagSNPs needed for each block is usually modest in the situation. The exhaustive searching algorithm enumerates next γ -combination in lexicographic order to generate the next candidate tagSNP loci set until each pattern can be uniquely distinguish.

D. Hadoop MapReduce framework

Hadoop is a software framework for coordinating computing nodes to process distributed data in parallel. Hadoop adopts the map/reduce parallel programming model, to develop parallel computing applications. The standard map/reduce mechanism has been applied in many successful Cloud computing service providers, such as Yahoo, Amazon EC2, IBM, Google and so on. An application developed by Map/Reduce is composed of Map stage and Reduce stage (optionally). Figure 2 illustrates the Map/Reduce framework. Input data will be split into smaller chunks corresponding to the number of Maps. Output of Map stage has the format of $\langle key, value \rangle$ pairs. Output from all Map nodes, $\langle key, value \rangle$ pairs, are classified by *key* before being distributed to Reduce stage. Reduce stage combines *value* by *key*. Output of Reduce stage are $\langle key, value \rangle$ pairs where each *key* is unique.

Hadoop cluster includes a single master and multiple slave nodes. The master node consists of a *jobtracker*, *tasktracker*, *namenode*, and *datanode*. A slave node, as computing node, consists of a *datanode* and *tasktracker*. Figure 3 illustrates the Hadoop cluster architecture. The jobtracker is the service within Hadoop that farms out Map/Reduce tasks to specific nodes in the cluster, ideally the nodes that have the data, or at least are in the same rack. A tasktracker is a node in the cluster that accepts tasks; Map, Reduce and Shuffle operations from a jobtracker.

Hadoop Distributed File System (HDFS) is the primary file system used by Hadoop framework. Each input file is split into data blocks that are distributed on datanodes. Hadoop also creates multiple replicas of data blocks and distributes them on datanodes throughout a cluster to enable reliable, extremely rapid computations. The namenode serves as both a directory namespace manager and a node metadata manager for the HDFS. There is a single namenode

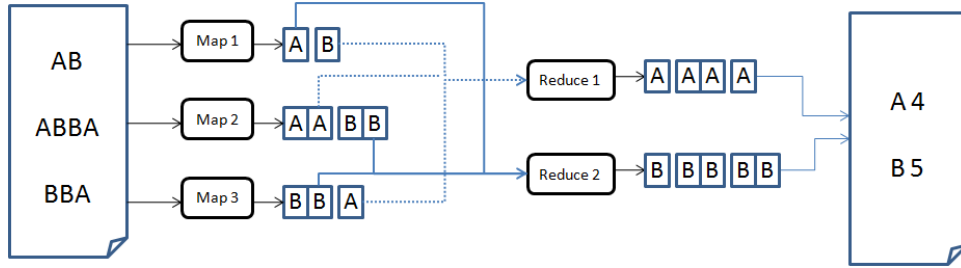


Figure 2. Example of counting words on MapReduce framework. The input data set $\{(A, B), (A, B, B, A), (B, B, A)\}$ are split into three maps, and each map process its data. Reduce process the data with same key. The output is number of “A” and “B”.

running in the HDFS architecture.

E. Haplotype block partitioning and selection on Map/Reduce framework

Figure 4 illustrates the MapReduce framework for the block partitioning and selection scheme. Assume that the number of map operations is N and the pattern length is L , the input $N \times L$ haplotype matrix is split into L/N chunks. Each map calculates the diversity scores of each block within the chunk where the map operation is responsible. Thus the output $\langle \text{key}, \text{value} \rangle$ pairs for each Map are $\langle (\text{block start number}, \text{block end number}), \text{diversity score} \rangle$ pairs. The map _{i} calculates diversity scores of blocks $\{\delta(i \cdot N/L, i \cdot N/L), \delta(i \cdot N/L, i \cdot N/L+1), \dots, \delta(i \cdot N/L+N/L, i \cdot N/L+N/L)\}$.

Therefore, each map has $(N/L)^2$ diversity scores. Reduce stage performs haplotype block selection algorithm. In our algorithm, just one reduce operation is needed in the reduce stage. Since the selection is a linear time algorithm, it is not necessary to perform the computation in parallel. The haplotype block selection algorithm is described in the previous section 2.A. The reduce operation finds the longest block by merging blocks with the interesting diversity scores.

3. Experiment

A. Experimental environment and data source

All of the experiments were performed on two IBM blade servers within our Cloud Computation laboratory. Each server is equipped with two Quad-Core Intel Xeon 2.26GHz CPU, 24G RAM, and 296G hard disk running under the operation system Ubuntu version 10.4 with Hadoop version 0.2 MapReduce platform. Under the current system environment, we control the server execution processes by up to 8 map operations and 8 reduce operations and the total number of the map/reduce operations are up to 16 respectively.

The SNP haplotype data sources are gathered from the International HapMap Project (<http://hapmap.ncbi.nlm.nih.gov/>),

which is a multi-country effort in order to identify and catalog genetic similarities and differences in human beings. This project collects many genetic sequences of different individuals. We downloaded the sequence data (Chromosome 1) from the HapMap3 Genome Browser release #2-African ancestry in Southwest USA (ASW). ASW includes 136 Chromosome 1 (chr 1) sequences (patterns) and the length of SNP is 116,416. These sequences are treated as the input data for our experiments. In the experiments, we applied formula (1) to calculate diversity scores of blocks.

B. Experimental results

To assess the performance of the proposed Hadoop MapReduce algorithm, we compare the computational time between various sequence data and various number of map/reduce operations. The sequential algorithm [23] has been proved as an efficient algorithm than other algorithms. Two factors, number of patterns and the length of patterns, affect the performance of sequential algorithm and the proposed algorithm. Patil *et al* [9] proposed that the haplotype block can be found within 300bp and 500bp. Therefore, the block size can be 300bp and 500bp. The diversity scores are calculated according to their corresponding block sizes; these scores are $\{\delta(1, 1), \delta(1, 2), \dots, \delta(1, 500), \delta(2, 2), \dots, \delta(2, 501), \delta(3, 3), \dots, \delta(L, L)\}$. Figures 5 and 6 illustrate the comparisons between sequential algorithm and our proposed algorithm under the MapReduce framework. In Fig. 5 and Fig.6, the block sizes are 300bp and 500bp, respectively.

The experimental results reveal that the computational time is effectively reduced when more map operations are deployed. Two and four map operations almost improve the computation time by factors of two and four times accordingly, comparing to the original sequential algorithm, respectively. Moderate enhancements between 8 and 16 map operations are observed in all experiments, since the size of data set split by 8 is similar to that by 16. Figure 7 illustrates the computation efficiency can be effectively improved proportional to the number of processors being used.

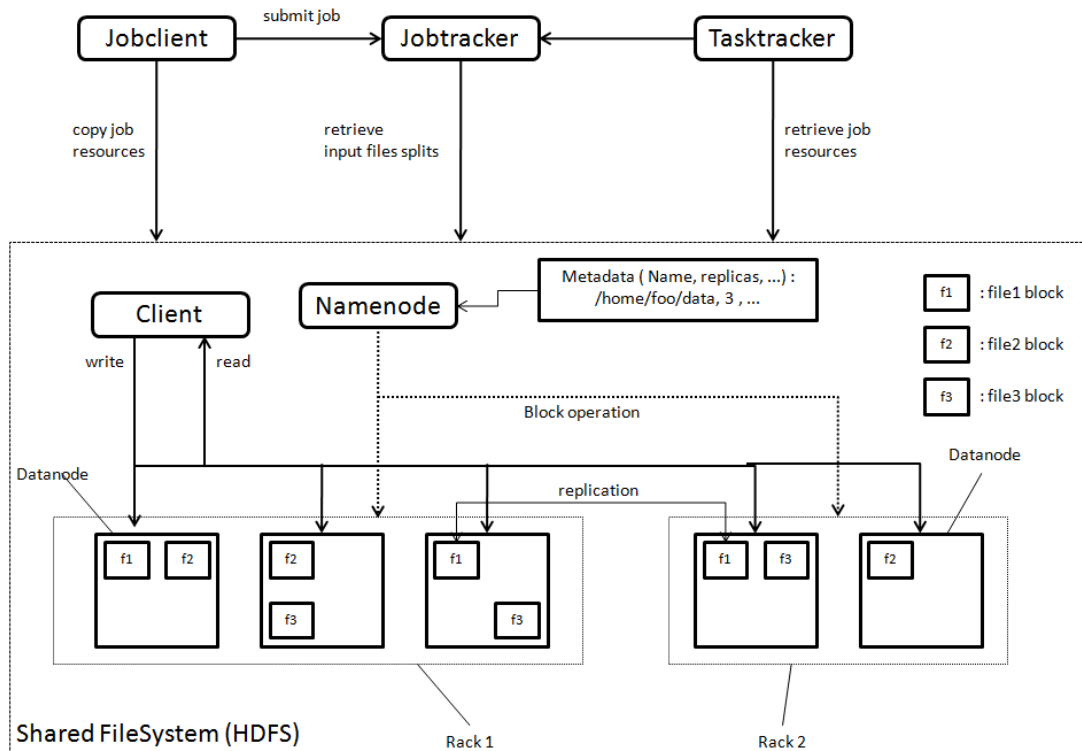


Figure 3. Hadoop Cluster architecture.

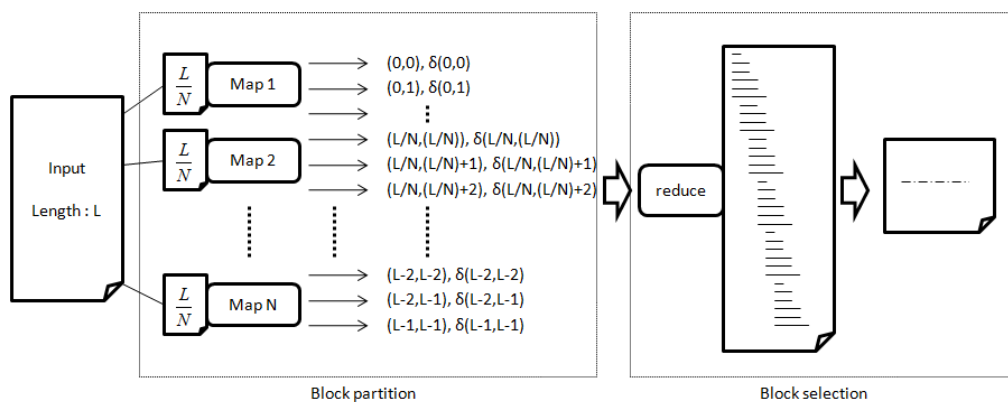


Figure 4. Haplotype block partitioning and selection on MapReduce framework.

In Fig. 5 and Fig. 6, it is observed that computational time increases corresponding to number of pattern and sequence length. The computational time for our algorithm with block size 300bp is less than that with block size 500bp. More patterns and longer sequence length lead to higher computational cost. These experimental results are corresponding to the algorithm analysis in previous section.

4. Conclusion

Studying on SNP and haplotype blocks can assist biomedical researchers to detect inherited diseases and contribute to classify the race of human and researching on

species evolution. The traditional methods for detecting haplotype blocks are based on dynamic programming approach. With the abundance of bioinformatic data that are all too common these days, the time-consuming traditional sequential methods require imminent assistance of the emerging parallel processing methodology. Here in this paper we discuss how we develop the parallelized frameworks improving our original dynamic programming algorithms, based on Hadoop map/reduce framework. The haplotype block partitioning copes with the problem of finding the minimum number of representative SNPs required to account for most of the haplotype block quality in each block. Due to the fault tolerance of Hadoop, the jobs

are just re-submitted to other nodes if the node is failure. This property is useful for analyzing large amount of sequence data since the job will not be stopped by node's fail. The experimental results show that the proposed algorithm can decrease the computational cost significantly.

In this paper, we compared the performance between various sequence lengths and pattern numbers. We also compared the performance between different block sizes. In the future, we will apply more diversity functions to the parallel algorithms to provide more perspectives for biologists to analyze these SNP data and investigate the relations of haplotype block selection between various block sizes and diversity measurements.

Acknowledgment

The authors would like to thank Siou-Hua Chang and Cheng En Hsieh for helping to perform experiments. This research was partially supported by the National Science Council under the Grants NSC-99-2632-E-126-001-MY3.

References

1. Bonnen, P.E., Wang, P.J., Kimmel, M., Chakraborty, R., Nelson, D.L., 2002. Haplotype and linkage disequilibrium architecture for human cancer-associated genes. *Genome Res.* 12, 1846–1853.
2. Gray, I.C., Campbell, D.A., Spurr, N.K., 2000. Single nucleotide polymorphisms as tools in human genetics. *Hum. Mol. Genet.* 9, 2403–2408.
3. Indap, A.R., Marth, G.T., Struble, C.A., Tonellato, P.J., Olivier, M., 2005. Analysis of concordance of different haplotype block partitioning algorithms haplotype tagging for the identification of common disease genes. *BMC Bioinformatics* 6, 303.
4. Mas, A., Blanco, E., Monux, G., Urcelay, E., Serrano, F.J., de la Concha, E.G., Martinez, A., 2005. DRB1-TNF-alpha-TNF-beta haplotype is strongly associated with severe aortoiliac occlusive disease, a clinical form of atherosclerosis. *Hum. Immunol.* 66, 1062–1067.
5. Nowotny, P., Kwon, J.M., Goate, A.M., 2001. SNP analysis to dissect human traits. *Curr. Opin. Neurobiol.* 11, 637–641.
6. Reif, A., Herterich, S., Strobel, A., Ehli, A.C., Saur, D., Jacob, C.P., Wienker, T., Topner, T., Fritzen, S., Walter, U., Schmitt, A., Fallgatter, A.J., Lesch, K.P., 2006. A neuronal nitric oxide synthase (NOS-I) haplotype associated with schizophrenia modifies prefrontal cortex function. *Mol. Psychiatry* 11, 286–300.
7. Daly, M.J., Rioux, J.D., Schaffner, S.F., Hudson, T.J., Lander, E.S., 2001. High-resolution haplotype structure in the human genome. *Nat. Genet.* 29, 229–232.
8. Gabriel, S.B., Schaffner, S.F., Nguyen, H., Moore, J.M., Roy, J., Blumenstiel, B., Higgins, J., DeFelice, M., Lochner, A., Faggart, M., Liu-Cordero, S.N., Rotimi, C., Adeyemo, A., Cooper, R., Ward, R., Lander, E.S., Daly, M.J., Altshuler, D., 2002. The structure of haplotype blocks in the human genome. *Science* 296, 2225–2229.
9. Patil, N., Berno, A.J., Hinds, D.A., Barrett, W.A., Doshi, J.M., Hacker, C.R., Kautzer, C.R., Lee, D.H., Marjoribanks, C., McDonough, D.P., Nguyen, B.T.N., Norris, M.C., Sheehan, J.B., Shen, N.P., Stern, D., Stokowski, R.P., Thomas, D.J., Trulson, M.O., Vyas, K.R., Frazer, K.A., Fodor, S.P.A., Cox, D.R., 2001. Blocks of limited haplotype diversity revealed by high-resolution scanning of human chromosome 21. *Science* 294, 1719–1723.
10. Dawson, E., Abecasis, G.R., Bumpstead, S., Chen, Y., Hunt, S., Beare, D.M., Pabial, J., Dibbling, T., Tinsley, E., Kirby, S., 2002. First-generation linkage disequilibrium map of human chromosome 22. *Nature* 418, 544–548.
11. Mahdevar, G., Zahiri, J., Sadeghi, M., Nowzari-Dalini, A., Ahrabian, H., (2010). Tag SNP selection via a genetic algorithm. *J. Biomed. Inf.*, doi:10.1016/j.jbi.2010.05.011.
12. Zhang, K., Calabrese, P., Nordborg, M., Sun, F., 2002. Haplotype block structure and its applications to association studies: power and study designs. *Am. J. Hum. Genet.* 71, 1386–1394.
13. Wall, J.D., Pritchard, J.K., 2003. Assessing the performance of the haplotype block model of linkage disequilibrium. *Am. J. Hum. Genet.* 73, 502–515.
14. Johnson, G.C.L., Esposito, L., Barratt, B.J., Smith, A.N., Heward, J., Di Genova, G., Ueda, H., Cordell, H.J., Eaves, I.A., Dudbridge, F., Twells, R.C.J., Payne, F., Hughes, W., Nutland, S., Stevens, H., Carr, P., Tuomilehto-Wolf, E., Tuomilehto, J., Gough, S.C.L., Clayton, D.G., Todd, J.A., 2001. Haplotype tagging for the identification of common disease genes. *Nat. Genet.* 29, 233–237.
15. Zahiri, J., Mahdevar, G., Nowzari-dalini, A., Ahrabian, H., Sadeghi, M., 2010. A novel efficient dynamic programming algorithm for haplotype block partitioning. *J. Theor. Biol.* 267, 164–170.
16. Greenspan, G., Geiger, D., 2004. High density linkage disequilibrium mapping using models of haplotype block variation. *Bioinformatics* 20, i137.
17. Wang, N., Akey, J.M., Zhang, K., Chakraborty, R., Jin, L., 2002. Distribution of recombination crossovers and the origin of haplotype blocks: the interplay of population history, recombination, and mutation. *Am. J. Hum. Genet.* 71, 1227–1234.
18. Hudson, R.R., Kaplan, N.L., 1985. Statistical properties of the number of recombination events in the history of a sample of DNA sequences. *Genetics* 111, 147–164.
19. Hadoop - Apache Software Foundation project home page [http://hadoop.apache.org/]
20. Taylor, R.C., 2010. An overview of the Hadoop/MapReduce/HBase framework and its current applications in bioinformatics. *BMC Bioinformatics*, 11:S1.
21. Dean, J., Ghemawat, S., 2010. MapReduce: A Flexible Data Processing Tool. *Communications of the ACM*, 53, 72–77.
22. Schatz, M., 2009. Cloudburst: highly sensitive read mapping with MapReduce. *Bioinformatics*, 25, 1363–1369.
23. Lin, Y.L., 2008. "Efficient Algorithms for SNP Haplotype Block Selection Problems." The 14th Annual International Computing and Combinatorics Conference (COCOON'2008), Springer Verlag, Lecture Notes in Computer Science 5092, 309–318.
24. Clayton, D., 2001. Choosing a set of haplotype tagging SNPs from a larger set of diallelic loci. *Nature Genetics*, 29(2).

25. Zhang, K., Qin, Z., Liu, J.S., Chen, T., Waterman, M.S., Sun, F., 2004. Haplotype block partitioning and tag SNP selection using genotype data and their applications to association studies. *Genome Res.* 14, 908–916.
26. Anderson E.C., Novembre, J., 2003, Finding haplotype block boundaries by using the minimum-description-length principle. *Am. J. of Human Genetics*, 73:336-354.
27. Li, W.H., Graur, D., 1991, *Fundamentals of Molecular Evolution*. Sinauer Associates, Inc.
28. Chapman, J.M., Cooper, J.D., Todd, J.A., Clayton, D.G., 2003, Detecting disease associations due to linkage disequilibrium using haplotype tags: a class of tests and the determinants of statistical power. *Hum. Hered.*, 56, 18-31.

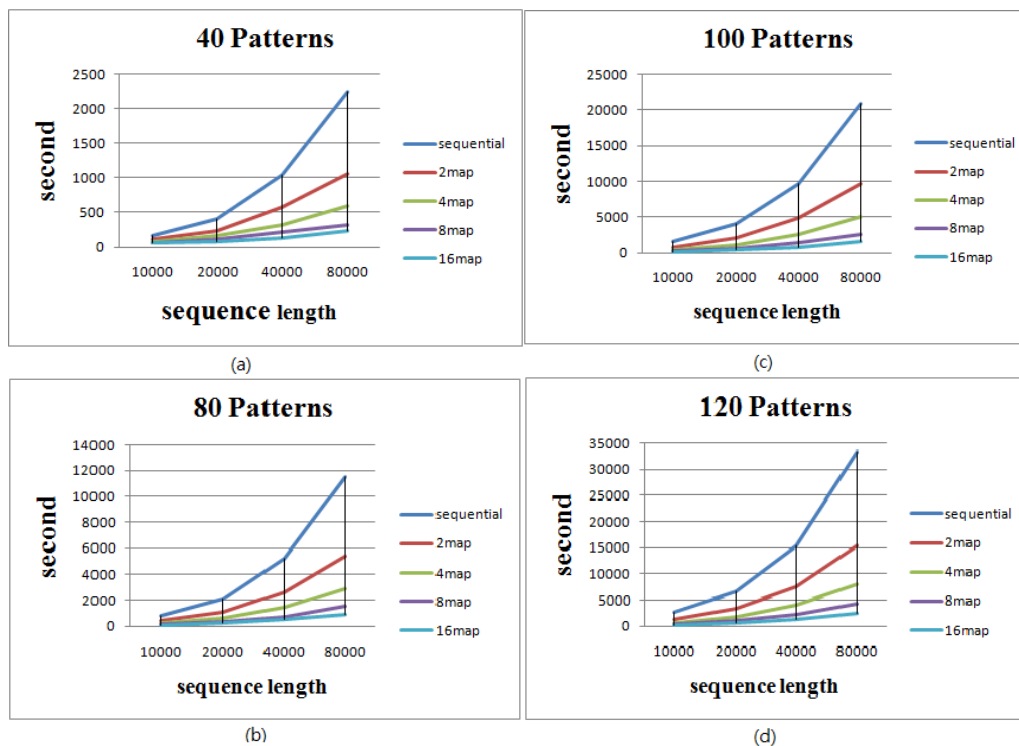


Figure 5. Performance comparison between sequential haplotype block selection and MapReduce haplotype block selection with block size 300bp.

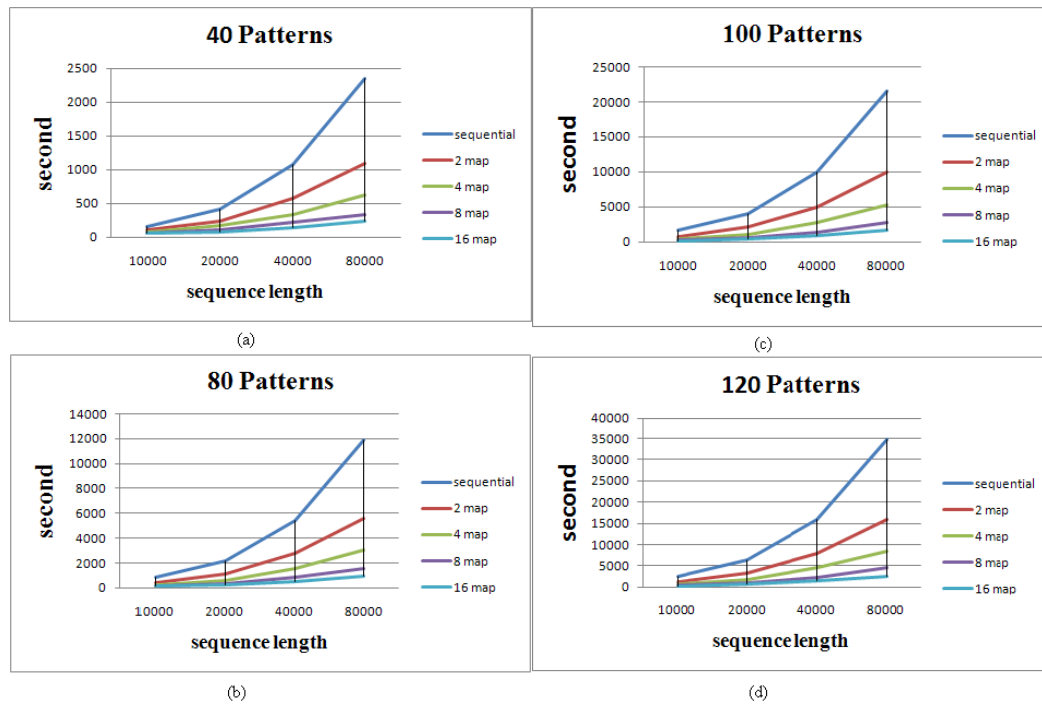


Figure 6. Performance comparison between sequential haplotype block selection and MapReduce haplotype block selection with block size 500bp.

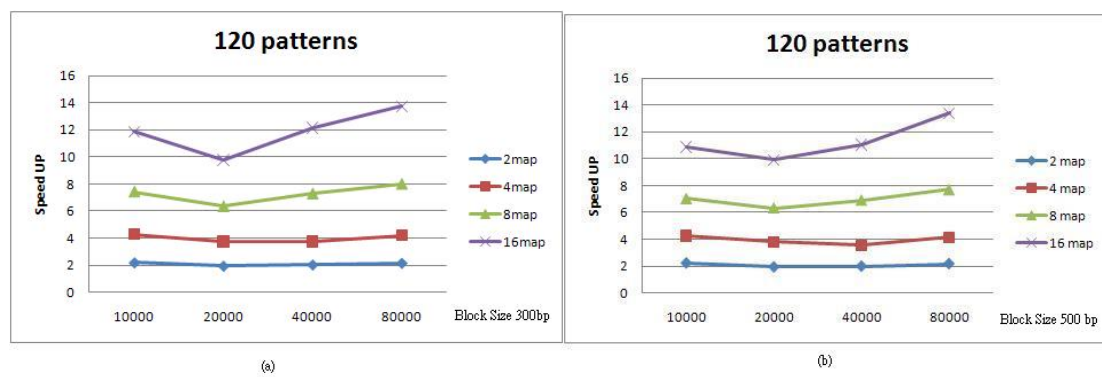


Figure 7. Speed up comparisons for MapReduce haplotype block selection over sequential haplotype block selection. (a) illustrates the speed up with block size 300bp. (b) illustrates the speed up with block size 500 bp.